

Karthik Bhattaram

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EDUCATION

University of California, Santa Barbara

Santa Barbara

Integrated B.S./M.S. in Computer Science

Sept. 2022 to June 2027

- Courses: Advanced Topics in Internet Computing, Machine Learning for Networking, Computer Security
- Honors: Regents Scholar, College of Engineering Honors, USACO Silver Division

EXPERIENCE

Software Engineer Intern

June 2025 – Sept. 2025

Arista Networks

Santa Clara, California

- Accelerated network switch memory diagnostics by orders of magnitude using custom C++ utilities
- Developed an **interprocess communication** tool for Microsoft and OpenAI datacenters

Machine Learning Research Intern

Sept. 2024 – June 2025

University of California, Santa Barbara

Santa Barbara, California

- Published a paper titled “Leveraging Large Language Models to Contextualize Network Measurements”
- Aggregated FCC data into a SQL database to improve model context using **retrieval-augmented generation**

Software Engineer Intern

July 2024 – Sept. 2024

Cadence Design Systems / OpenEye Scientific

Santa Fe, New Mexico

- Wrote **AWS**-based package to benchmark database compression algorithms that saved terabytes of server space
- Updated **CUDA** C++ software to incorporate new language features to match competing industry solutions

Hardware Architecture Intern

June 2023 – Sept. 2023

Cadence Design Systems

San Jose, California

- Implemented an automation tool that generated memory wrappers for use on a hardware emulation platform
- Collected **System Verilog** files to be used as templates for this tool and tested memories it produced

Software Engineer Intern

June 2021 – Aug. 2022

Axiado Corporation

San Jose, California

- Developed a package to create **network packet flows** based on IPv6, ARP, ICMP, etc. in simulation
- Created and executed a test plan for a hardware firewall’s parser module and discovered multiple bugs

PROJECTS

Graph Fine Tuning | *Python, PyTorch, Graph Algorithms*

Jan 2026 – March 2026

- Adapted Meta’s Llama 3.1 model to process graph generation tasks using parameter-efficient fine tuning
- Implemented an experimental pipeline to compare zero-shot, Supervised Fine Tuning, and Direct Preference Optimization performance
- Achieved 85% improvement in graph correctness (based on graph metrics) over the zero shot model

College Course Database | *Java, SQL, Oracle Cloud*

March – May 2025

- Designed a database schema using entity relationship diagrams
- Wrote SQL commands to create the tables and enforce data integrity
- Implemented a Java backend with JDBC to manage database transactions
- Created command line interfaces for the registrar and students with distinct permissions

TECHNICAL SKILLS

Certificates: Enterprise Security Professional, Python for Data Science and AI

Languages: Python, C/C++, Java, JavaScript, Shell, Verilog

Developer Tools: Git, Visual Studio Code, Vim, Emacs

Libraries: CUDA, Pandas, NumPy, PyTest, Black, Matplotlib